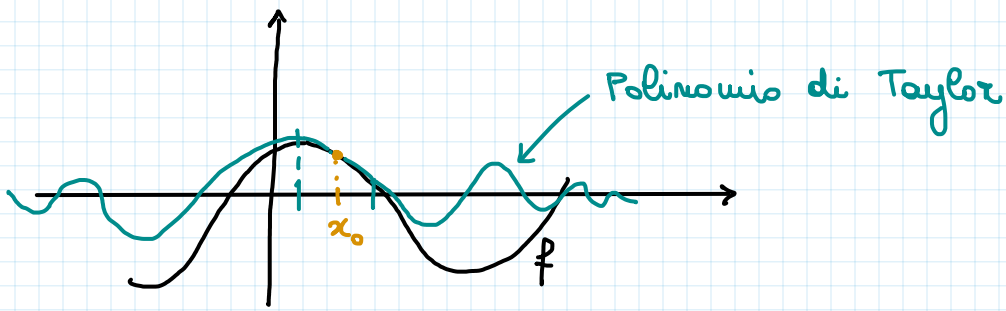

TEOREMA (SVILUPPO in SERIE di TAYLOR) $N=1$

Se $f: \mathbb{R} \to \mathbb{R}$ di ordine $\mathcal{C}^k(\mathbb{R})$, sia $x_0 \in \mathbb{R}$ e $x \in B(x_0, r)$ $\quad r>0 \quad r \in \mathbb{R}$

Allora:

$$f(x) = \boxed{\sum_{i=0}^{k} \frac{f^{(i)}(x_0)}{i!} \cdot (x-x_0)^i} + \overbrace{R_k(x) \cdot (x-x_0)^k}^{\text{RESTO}}$$

$$\lim_{x \to x_0} R_k(x) = 0$$

Polinomio di Taylor

ESEMPIO $N=2$

$f: \mathbb{R}^2 \to \mathbb{R} \qquad f(x,y) = x^2 + ye^x$

$$\frac{\partial^1 f}{\partial x^1} = 2x + ye^x \qquad (1,0) \qquad\qquad \frac{\partial^1 f}{\partial y^1} = e^x \qquad (0,1)$$

$$\frac{\partial^2 f}{\partial x^2} = 2 + ye^x \qquad (2,0) \qquad\qquad \frac{\partial^2 f}{\partial y^2} = 0 \qquad (0,2)$$

$$\frac{\partial^{1+1} f}{\partial x^1 \partial y^1} = e^x \qquad (1,1)$$

MULTI-INDICI

Def: Sia $f: \mathbb{R}^N \to \mathbb{R}$

$$\partial^\alpha f = \frac{\partial^{\alpha_1+\alpha_2+\dots+\alpha_N} f}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \partial x_3^{\alpha_3} \cdot \partial x_N^{\alpha_N}}$$

α MULTI-INDICE

$= (\alpha_1, \alpha_2, \dots, \alpha_N)$

$|\alpha| = \alpha_1 + \alpha_2 + \dots + \alpha_N$